




A Joint Project of the Manitha Neyam Trust, Hindu Educational Society & The All Ceylon Hindu Congress to Improve the Achievement Level of the G.C.E(A/L) Students in Mathematics, Biology & Commerce - 2026
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Biology -I
Multiple Questions
2hours

Answers all Questions

01. Hexose is present as monosaccharide component in
1) Pectin 2) Chitin 3) Inulin 4) Hemicellulose 5) Cellulose
02. Which of the following statements regarding the basic features shared by all cells are **correct**?"
a. Bounded by a plasma membrane
b. Carry RNA as genetic material
c. Ribosomes are found
d. fluid cytoplasm
1) a and b 2) a and c 3) a and d 4) ☐ b and c 5) b and d
02. Hexose is present as monosaccharide component in
1) Pectin 2) Chitin 3) Inulin 4) Hemicellulose 5) Cellulose
03. Select the **correct** statement regarding the Golgi apparatus
1) It produces endoplasmic reticulum
2) Trans face is located near the endoplasmic reticulum
3) Modifying materials
4) Manufacturing cellulose only cell wall components
5) Network of tubular sacs structure
;
04. Which of the following is **correct** regarding the microscope
A – Resolution power of the light microscope is a measure of the clarity of the image
B – In practice electron microscope magnifies about 1×10^5 with a resolution of 0.2nm
C – In a transmission electron microscope electrons do not pass through the specimen but reflect
D – In a scanning electron microscope all the electrons are scattered and are never absorbed by the specimen
The **correct** statements from above are.
1) A only 2) A and B only 3) A, B and C only 4) A and C only 5) A, C and D
05. Select the **correct** statement about the extracellular matrix of animal cells.
1) Forms a protective layer over the cell surface
2) Main components are glycoproteins and carbohydrates only
3) A layer of proteoglycan secreted by cells is present
4) Collagen linking only extracellular matrix
5) Most abundant glycoprotein in the extracellular matrix of all animal cells is collagen
06. **Minimum** contribution for enzyme synthesis is given by
A – Endoplasmic reticulum B – DNA C – Ribosome D – mRNA E- rRNA
1) A and B 2) A and C 3) A and D 4) A only 5) B only

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07. Select the **correct** answer based on the following two statements

A - C₄ pathway helps plants to improve the efficiency of CO₂ fixation at lower CO₂ concentrations.

B - CO₂ fixation occurs twice, first in mesophyll cells, and second in bundle sheath cells.

- 1) Both A and B are correct, and B contributes to A
- 2) Both A and B are correct, and B does not contribute to A -
- 3) A is correct and B is incorrect
- 4) A is incorrect and B is correct
- 5) Both A and B are incorrect

08. Some statements regarding the diversification of eukaryotes are given

- a) Ancestors of arthropods and chordates originated
- b) Sponges evolved
- c) Colonization of land by fungi plants and animal
- d) Differentiation of large tree as roots stem and leaves

The **correct** sequence of diversification is,

- 1) a,b,c,d
- 2) b,c,a,d
- 3) b,a,c,d
- 4) b,d,c,a
- 5) b,c,d,a

09. Which of the following responses is/are **correct**

A; Branched hydrocarbons in membrane lipids are *Thermococcus and Salmonella*

B; Some lycophytes are heterosporous

C: Fungi cell walls are made up of cutin

- 1) A only
- 2) B only
- 3) C only
- 4) A and B only
- 5) A and C only

10. Select the **correct** statement regarding the evolution of vascular plants

- 1) Lycophyte plants have a more recent common ancestor with seed plants
- 2) Colonization of land by seedless vascular plants began after about 670 million years ago
- 3) Few seedless vascular plant species produce one type of sporangium and one type of spores
- 4) Modified leaves that bear sporangia are known as sporophylls
- 5) Fossils suggest that ancestors of vascular plants had gametophytes and sporophytes that were unequal in size

11. Which of the following classes include/includes animals having scales

a- Chondrichthyes

b- Osteichthyes

c- Amphibia

d- Reptilia

e- Aves

f- Mammalia

- 1) a,b,d only
- 2) a,b,c only
- 3) a,c,f only
- 4) a,b,d,e only
- 5) a,d,e,f only

12. Select the **correct** statement regarding the plant tissue system

- 1) Mature parenchyma cells have a primary cell wall and a secondary cell wall
- 2) Few of the parenchyma cells retain the ability to divide and differentiate under suitable conditions
- 3) Collenchyma cells have thicker primary walls than parenchyma cells
- 4) Vessel elements and tracheids are present in all vascular plants
- 5) End walls of the sieve tube elements are completely broken down to allow movements of fluid

13. During the opening of stomata

A – Guard cells actively accumulate K⁺ from neighboring epidermal cells

B – Exosmosis of water from guard cells

C – Turgor pressure in guard cells decrease

D – Abscissic acid also plays a role

Select the **correct** response

- 1) A only
- 2) A and B only
- 3) A and C only
- 4) A and D only
- 5) B, C and D only

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14 Select the **correct** statement.

- 1) Only heterostyly is a typical characteristic of flowers for cross-pollination
- 2) In double fertilization one sperm nucleus fuses with two synergids
- 3) Gametophyte of angiosperm is reduced structure with eight cells
- 4) Seed contains the embryo and stored food, surrounded by the seed coat
- 5) Parthenocarpy occurs naturally in orange

15. Some deficiency symptoms in plants are given below

- a) **Poor flowering**
- b) **Death of terminal buds**
- c) **Reduced in internodes length**
- d) **Stubby roots**

Their responses with the **correct** sequence of elements responsible for the above symptoms are,

- 1) Ca, Zn, Cu, P 2) P, Ca, Zn, Cu 3) P, Ca, Zn, Cl 4) Zn, Ca, Cl, P 5) P, Zn, Ca, Cl

16. In which event of the plant life cycle does **meiosis** occur.

- 1) During a single egg is produced within the archegonium of *Pogonatum*
- 2) During the division of the zygote of *Nephrolepis*
- 3) During megasporangium produce four large megaspores of *Selaginella*
- 4) During many archegonium produces a single egg of *Cycas*
- 5) During production of the two sperm of *Cocos*

17. Select the **correct** statement regarding blood tissue

- 1) Specialized connective tissue where the matrix is secreted by the blood cells
- 2) Fibres are formed only during the blood clotting process
- 3) Extracellular matrix of the blood is semisolid
- 4) Red blood cells and white blood cells involve in defense
- 5) Main function of blood tissue is transport of material only

18. The ECG tracing of a healthy individual is given below

A – P wave – Represents the impulse from the SA node as it sweeps over the atrial depolarization

B – QRS wave – Spread of impulse from the AV node throughout ventricular repolarization

C – T wave – Ventricular repolarization and relaxation of the ventricular muscles

Which of the following combinations is/are **correct**?

- 1) A only 2) B only 3) A and B only 4) A and C only 5) B and C only

19. Select the **correct** answer based on the following three statements

A – Blood pH falls due to rising levels of CO₂ in tissue during exercise

B – Medulla oblongata receives signals from blood capillaries

C – Signal from medulla oblongata to respiratory muscle increases the rate and depth of ventilation

- 1) Both A and B 2) Both A and C are correct 3) Both B and C are correct
- 4) A is incorrect and B is correct 5) Both A and B are incorrect

20. Several statements regarding blood group are given below

A – Person having blood group O⁻ Cannot receive any different type of blood group

B – Mother having blood group AB⁻ conceived with blood group AB⁺ fetus does not cause any complication

C – Person having blood group AB⁺ can receive blood from any type of blood group

Which of the above statements are **correct**

- 1) A and B only 2) A and C only 3) B and C only 4) A, B and C 5) A only

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21. Few statements regarding the nervous system of animals are given below

A- Cnidaria – have nerve net

B- Annelida – have brain and dorsal nerve cords

C- Echinodermata – have nerve ring and radial nerve

D- Chordata – have brain and dorsal nerve cord

Correct from above

- 1) A and B only 2) A and C only 3) A and D only 4) A, C and D only 5) A, B and D only

22. **Correct** statement regarding the sensory receptors and function of them is,

- 1) Rods – Distinguish the colour of red, and green mainly at night
- 2) Retina neurons – One type of neuron present including bipolar cell
- 3) Lens – Elastic biconcave transparent disc made up of protein transparent capsule
- 4) Ear ossicle – Extend across the middle ear tympanic membrane to the round window
- 5) Semicircular canals – Contribute to maintain the posture and balance

23. Which is **correct** regarding site of production and function of the following hormones

Hormone	Site of production	Function
1) PRH	: Anterior pituitary	: stimulates and secretion of prolactin
2) Growth Hormone	: Hypothalamus	: promotes tissue growth
3) TRH	: Anterior pituitary	: stimulates growth of thyroid gland
4) ACTH	: Anterior pituitary	: stimulates secretion of glucocorticoid hormone
5) LH	: Hypothalamus	: stimulate ovulation

24. Sexual reproduction is different from asexual reproduction as **sexual reproduction**,

- 1) Allow to clear favorable phenotype which would enhance reproductive success
- 2) Beneficial gene combination arising through recombination may speed up adaptation
- 3) Allow rapid multiplication of individuals from single-parent
- 4) Mitosis recombination during help produces varied genotypes
- 5) Produces offspring which are genetically identical to each other

25. Human semen contains the following components

- | | | |
|---------------|-----------------|------------------|
| A Basic mucus | B Enzyme | C Prostaglandins |
| D Citrate | E Ascorbic acid | |

Which component added to semen from the prostate gland

- 1). Cand E only 2). A and B only 3). A and D only 4). Band D only 5). A and E only

26. Few statements regarding the birth control methods of human are given below

a- oral contraceptive for female - has high concentration of progesterone only

b- condoms for male - barrier device which prevents sperm entry

c- IUD (loop) for female - prevent implantation of a fertilized ovum

d- Depo provera injection for female - prevents implantation by making endometrium thick

which are **correct** from the above methods

- 1) a and c only 2) b and c only 3) b and d only
4) a and d only 5) a and b only

27. A few symptoms of sexually transmitted infections are given below

- Skin rashes
- Sores on vagina
- Fever

The above symptoms could be an indication of

- 1) Gonorrhea 2) Syphilis 3) AIDs 4) Genital herpes 5) Gonorrhea and Syphilis

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28. Which answer contains bone as pairs only in the human skull

- 1) Maxilla and Mandible 2) Temporal and Ethmoid 3) Vomer and Zygomatic
4) Frontal and Temporal 5) Temporal and Parietal

29. A) The bones of the foot are arranged in arches

B) Femur is the strongest bone

C) Knee is locked when it is fully extended

which arrangement is/are the adaptation for human lower limb bearing body weight when walking

- 1) A only 2) B only 3) C only 4) A and B only 5) B and C only

30. Select the **correct** statement regarding disorders of human skeleton system

- 1) osteoporosis is a degenerative non inflammatory disease
2) Osteoarthritis is reduction of bone density
3) Slipped disc may cause numbness and discomfort
4) Gender, aging, heredity and obesity are risk factors contribute for osteoporosis
5) Factors causing osteoarthritis includes hormonal imbalances

31. Select the **correct** combination regarding human Mendelian characteristics

A – Attached earlobes-Recessive

B – Absence of dimples-Dominant trait

C – Tongue rolling ability-Recessive trait

D - Show Widows peak-Dominant trait

- 1). A and C only 2) A and D only 3) A and B only 4) B and D only 5) C and D only

32. Black Fruit fly (**B**) is dominant over brown colored (**b**), long wing (**W**) is dominant to vestigial wing (**w**). Fly which is heterozygous for both characters is crossed with the **BbWW** genotype. What is the probability of obtaining brown-colored and long-wing fly in the progeny from the above cross?

- 1) 1/16 2) 1/8 3) 1/4 4) 3/8 5) 3/4

33. Select the statement which **isn't** correctly matched regarding mutations of human genome.

- 1). Klinefelter's syndrome - presence of an extra X chromosome.
2). Down's syndrome - absence of one X chromosome.
3). Turner syndrome - presence of X chromosome only
4). colour blindness - a recessive gene mutation on the X chromosome.
5). Hemophilia - a recessive mutation of a gene on the X chromosome.

34. Select the answer which contains first enzyme used for DNA replication and the most important enzyme for DNA replication

- 1.) Helicase and DNA polymerases. 2). RNA polymerases and DNA polymerases.
3). Topoisomerase and RNA polymerases. 4). Helicase and RNA polymerases.
5). Topoisomerase and DNA polymerases.

35. Select the **correct** combination

- A. Basel convention – Control of transboundary movement of hazardous wastes and their disposal
B. Ramsar – Prevention of drylands and their resources
C. Kyoto Protocol – Reducing greenhouse gas emissions
D. CITES convention – Protection of fauna and flora by establishment of national reserves

- 1) A and B only 2) A and C only 3) A and D only 4) A, B and D only 5) A, C and D only

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36. Select the **incorrect** combination

- 1). Invasive species – Giant African land snail
- 2). Endemic species – Lingula
- 3). Indigenous species – Kitul
- 4). Exotic species – Tilapia
- 5). Migratory species – Indian fly catcher.

37. Not a feature of prions

- 1). Size is smaller than the virus
- 2). Same like viruses, prions contain protein and nucleic acids
- 3). Replicate with the help of the host's gene.
- 4). Disease-causing agents in some birds and mammals
- 5). Human-to-human disease transmission is associated with transfusion of infected blood and tissue

38 Select the **incorrect** statement

- | | | |
|------------------------|---|----------------------|
| 1). Respiratory tract | - | Rabies virus |
| 2). Digestive tract | - | Hepatitis A virus |
| 3). Reproductive tract | - | Herpes simplex virus |
| 4). Skin | - | Measles virus |
| 5). Eye | - | Influenza virus |

39. Select the **correct** statement regarding the food spoilage

- A) Protein food is broken down by the proteolytic enzymes into ammonia and hydrogen sulfide only
 B) Carbohydrate is converted into only acid and alcohol
 C) Lipids are converted into fatty acids and glycerol
 D) Foodborne pathogens of humans is *Shigella* and *Vibrio cholera*
- 1) A and C only 2.) A and D only 3, A and B only 4 B and D only 5.) C and D only

40. Which of the following are **not** the general characteristics of aquaculture species?

- 1). It should grow well in prevailing physical and chemical parameters of water in the area
- 2). It should be easy to breed so that sufficient number of fertilized egg could be obtained easily
- 3). It should accept formulated food and grow well
- 4). If it reaches sexual maturation relatively early it is advantageous
5. It should not reproduce in grow-out ponds

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- For each of the questions 41 to 50 one or more of the responses is/are correct. Decide which response/responses is/are correct and then select the correct number.

If only A, B and D are correct 1
 If only A, C and D are correct 2
 If only A and B are correct 3
 If only C and D are correct 4
 If any other response or combination of responses is correct 5

Directions summarised				
1	2	3	4	5
A, B, D correct.	A, C, D correct.	A, B correct.	C, D correct.	Any other response or combination of responses correct.

41. Which of the following statements regarding photophosphorylation is/are **correct**
- A) Substrate level phosphorylation occurs in glycolysis and Krebs cycle in aerobic respiration
 B) Oxidative phosphorylation produces a higher number of ATPs from aerobic respiration
 C) NADP⁺ is used as an electron acceptor in both photophosphorylation and oxidative phosphorylation.
 D) Both photophosphorylation and oxidative phosphorylation occur only in the presence of light
 E) Photophosphorylation takes place in the grana of chloroplast while oxidative phosphorylation takes place in the stroma of chloroplast
42. Which of the following statement/statements is/are **correct** regarding the cork cambium and the tissue produced by it during the secondary growth of woody plants
- A) Cork cambium and tissues it produces are collectively called periderm
 B) Due to the deposition of suberin in cork cells it becomes impervious to water and air
 C) Lenticels are vertical slits in bark of the plant
 D) New cork cambium always forms interior to the cork cambium of root
 E) bark is the cork cambium and all tissues produced by it
43. Which of the following is/are a structure / structures within the seed which is / are formed as a result of fertilization of eggs in flowering plants
- A) Antipodal cells B) Integument C) Endosperm D) Seed coat E) Synergid cells
44. Which of the following structure/structures can be seen in at least two phyla
- A) Dorsal nerve cord B) Book lungs C) Gastrovascular cavity D) Haemocoel E) Malphigian tubule
45. Which of the following statements is/are **correct** regarding the sensory receptors
- A) A special structure designed to receive specific stimuli
 B) Stimulated by chemicals only
 C) Always connected with the nervous system
 D) If the stimulation is continuous many receptors show a decrease in responsiveness
 E) Detect the stimulus if the stimulus is at or lower than the threshold level

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46. By which character/s the human lymphatic system differs from circulatory system

- A) The lymphatic system bears no heart, but the circulatory system bears a heart
- B) Lymphatic vessels possess valves, but blood vessels do not possess valves
- C) Lymph flows only towards the heart but blood flows towards or away from the heart
- D) Lymphatic capillaries are blind ended while the blood capillaries are not blind
- E) Both lymph and blood have the same composition

47. Raw materials required for polymerase chain reaction is/are

- A) dNTPs B) Mg^{2+} C) Taq RNA polymerase D) Primer E) DNA ligase

48. Which statement/s regarding the main terrestrial biomes of the world is/are **not** agreeable

- A) Tropical rainforest - Annual rainfall between 1500 to 2000mm
- B) Chaparral – No tall trees, small trees, shrubs and herbaceous plants present
- C) Coniferous forests – Smallest biomes of the world, plants bear adaptations to avoid accumulation of snow
- D) Temperate grassland – Two types of grasses - tall grasses and mixed grasses
- E) Desert – Deep roots in shrubs

49. Which of the following statements is/are **correct** regarding antibiotics

- A) Many antibiotics are produced by microbial fermentation
- B) Only fungi are used for commercial production of antibiotics
- C) Tetracycline is an antibiotic which inhibits protein synthesis in bacteria
- D) Antibiotics are secondary metabolites of microorganisms
- E) Antibiotics affect against broad range of bacteria and viruses' broad-spectrum antibiotics

50. Select the **correct** statement regarding Dengue.

- A) Vector-borne disease caused by an arbovirus which is a RNA virus
- B) The infected humans are always symptomatic
- C) The dengue virus is transmitted to humans via the biting of an infected female mosquito
- D) Breeding sites are dark-colored surfaces with clear water
- E) The infected humans act only as the carriers of the virus